

Ravolexin Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-934/B

Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. The control strategy links critical quality attributes to critical process parameters identified during development. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions.

Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. The control strategy links critical quality attributes to critical process parameters identified during development. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation.

Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Cleaning verification swab results were below the health-based exposure limits derived for the product. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates

of suitability are maintained where applicable.

No changes to the approved analytical procedures are proposed in this submission. Cleaning verification swab results were below the health-based exposure limits derived for the product. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

Control of Excipients (3.2.P.4)

For the commercial formulation F-934/B, Kollidon 90F (Povidone, binder) is used.

Excipient Grade Selection

The basis for selection of Kollidon 90F is reduced lot-to-lot variability reported by the qualified supplier, demonstrated across three pilot-scale batches.

Povidone complies with the requirements of ICH Q8(R2), ICH Q3D and FDA Guidance for Industry: Immediate Release Solid Oral Dosage Forms.

Current Commercial Specification

The current approved commercial specification (SPEC-EX-2303 Rev 08) lists Povidone as Plasdone K-29/32.

Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. The control strategy links critical quality attributes to critical process parameters identified during development.

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Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. The control strategy links critical quality attributes to critical process parameters identified during development. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. The control strategy links critical quality attributes to critical process parameters identified during development.

Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

Deviations observed during the campaign were investigated and closed with no impact to product quality. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Statistical evaluation of

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Facilities and Equipment

Manufacturing is performed at the approved commercial site using dedicated product-contact equipment trains. Equipment qualification and cleaning validation are maintained under the site validation master plan. No changes to the approved facility registration are proposed.

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